The Modal–Dependence Calculus (MDC)

A Minimal Modal Framework for Invariance and Dependence

"What repeats is not the thing but the structure of dependence under invariant constraint."

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Abstract

A longstanding thesis in metaphysics is that contingent entities depend for their existence on something that cannot fail to exist. This paper develops the *Modal Dependence Calculus* (MDC), a minimal extension of quantified *S5* that formalizes this dependence using only modal resources and a world-relative existence predicate $E(x)$, where $E(x)$ indicates that x is actualized at a world w. Dependence is defined via asymmetric modal entailment: $D(x, y)$ holds if and only if x cannot be actualized unless y is, while y can be actualized without x.

Nine axioms (A0–A8) encode the structural constraints on dependence: irreflexivity, asymmetry, transitivity, the impossibility of infinite descent, and the linkage between invariance and necessity. From these, MDC establishes three central theorems: (i) every contingent entity depends on something invariant across possible worlds; (ii) any invariant entity is necessary within the adopted modal framework; and (iii) there is exactly one such invariant necessary terminus within that framework, called the *Core* (**c**).

MDC avoids modal collapse: contingent entities remain genuinely contingent, as guaranteed by A5. It thereby supplies a purely modal surrogate for several key features of grounding—directionality, priority, and essential dependence—without primitive hyperintensional operators. As a consequence, a structural form of monism emerges as a theorem rather than an assumption: in any non-empty world, the Core is actualized, and all invariant necessary termini coincide with it. MDC does not establish that monism is true; it shows that a form of structural monism is derivable under the adopted constraints.

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1 Introduction

Contingent entities could have failed to exist. If Socrates exists, his existence is not self-explanatory: something must account for why he exists rather than not. Traditional options—infinitary descent, explanatory cycles, and brute contingency—each face familiar difficulties. Infinite chains never resolve into an explanatory ground; cycles violate the asymmetry characteristic of explanation; and brute facts abandon the project of intelligibility.

A recurring theme in metaphysics is that dependence terminates in what is invariant. Aristotle's

unmoved mover exemplifies this idea: what varies depends on what does not. The invariants of modern physics exhibit a structurally similar role in a different domain. This paper develops a modal framework that captures this structure within the standard resources of quantified *S5*.

The *Modal–Dependence Calculus* (MDC) introduces a dependence operator defined as asymmetric modal entailment:

$$D(x, y) \;\equiv\; \Box(E(x) \to E(y)) \;\wedge\; \neg\Box(E(y) \to E(x)),$$

where $E(x)$ is a world-relative existence predicate. Intuitively, if x is actualized, y must be as well; but y may be actualized without x. This captures the directionality of metaphysical explanation without appealing to primitive grounding relations.

MDC is a conservative extension of standard *S5*: it adds no modal strength beyond the underlying logic. Its axioms A0–A8 impose structural constraints on dependence—irreflexivity, asymmetry, transitivity, well-foundedness, and two principles linking invariance with necessity. From these, the system derives three central results: (i) every contingent dependence chain terminates, (ii) its terminus is necessary, and (iii) it is unique.

The term *calculus* is used here to denote a systematic framework for deriving structural consequences, not a complete proof-theoretic deductive system. MDC is a model-theoretic framework; completeness, decidability, and proof-theoretic development remain future work.

The central novelty of MDC lies not merely in recovering explanatory termination, but in deriving the uniqueness of an invariant terminus within plain quantified *S5*, without invoking a primitive grounding operator or additional hyperintensional machinery. In this way, MDC offers a modal surrogate for the core features traditionally associated with grounding—directionality, priority, and essential dependence—while remaining formally conservative.

Just as importantly, MDC avoids modal collapse. In particular, axiom A5 ensures that $\Diamond\neg E(x)$ holds for every contingent x. The framework thus distinguishes modal necessity, which belongs exclusively to the Core, from the modal variability characteristic of dependent entities.

The remainder of the paper develops the formal theorems, presents explicit models, addresses major objections, and shows how classical claims of monism arise as consequences rather than assumptions. The final corollary states that if any world contains being, then the Core is actualized there, and any necessary invariant terminal entity coincides with it.

Methodological Clarification. MDC is a structural model framework for deriving conditional consequences about dependence, invariance, and termination. It does not assume a primitive grounding relation, but defines dependence purely modally and shows that, under the stated constraints, a unique invariant terminus plays the explanatory role traditionally assigned to metaphysical ground. The results impose structural constraints on any adequate account of contingency, while substantive interpretive questions about the nature of the Core lie beyond the calculus itself. MDC specifies

structural conditions any adequate account of dependence must respect; it does not itself determine the metaphysical interpretation of those conditions.

All results in MDC are conditional on the adopted structural constraints (A0–A8). The framework does not assert that these constraints must hold, but derives what follows if they do. Its contribution is therefore diagnostic: MDC clarifies which structural commitments a theory of dependence must accept, reject, or supplement if it aims at explanatory completion.

Scope and Limitations. The results of MDC are strictly structural and modal. The calculus does not determine the intrinsic nature of the Core, nor does it offer a causal, temporal, or agential account of dependence. Accordingly, the Core is to be understood as a formally characterized invariant terminus rather than a fully interpreted metaphysical entity. Any stronger ontological or theological commitments belong to the interpretation of the system, not to its formal results.

Alternative frameworks that reject well-founded dependence or deny that dependence is directed toward invariance fall outside the class of structures to which MDC applies.

Relation to Grounding and Monism. MDC is not a formalization of grounding in the hyperintensional sense developed by Fine, Litland, and others, but a modal surrogate capturing structural features such as asymmetry, directionality, priority, and termination.

Similarly, the emergence of a unique invariant terminus should not be read as the direct assertion of classical substance monism. Rather, it is a structural consequence of the adopted constraints: under constant-domain *S5* with A0–A8, any terminal invariant necessary entity is unique. In this sense, MDC derives a form of structural monism from modal dependence rather than assuming it.

Roadmap. Section 2 introduces the formal background used throughout the paper. Section 3 states the axioms with brief informal glosses. Section 4 proves the three central theorems: existence and uniqueness of the Core, termination, and modal conservation. Section 5 presents a simple three-world model illustrating consistency. Section 6 addresses major objections. Section 7 develops consequences for essence and origin. The appendices provide full proofs, extended models, and worked examples.

Invariance as Method. Relativity removes absolute simultaneity to clarify physical law; MDC removes absolute contingency to clarify ontological structure. The methodological move is analogous: identify a structural asymmetry in how a domain is described, replace absolute primitives with invariant constraints, and derive the consequences.

This parallel is methodological, not empirical. MDC makes no predictions and offers no causal account. What it shares with invariance-driven frameworks in physics and mathematics is the core commitment: that explanatory structure becomes clearest when anchored in what remains invariant under transformation. Importantly, the two operate at complementary levels: physical theories

employ invariance to describe *how* systems behave under transformation, taking invariant constraints as given; MDC operates at a different level, showing *why* any dependence structure that aims at explanatory completion must terminate in something invariant. Physics takes invariance as given where MDC shows it to be required. Under this reading, A6 functions not as an arbitrary assumption but as a constraint on intelligibility—the requirement that contingency be structurally oriented toward something that does not itself depend. The Core is not posited; it is what the invariance constraint requires.

2 Symbol Map and Dependence Schema

Symbol	**Meaning**
$\mathcal{D}$	fixed domain of possible individuals
$E(x)$	x is actualized at a world w (world-relative)
$C(x)$	x is contingent (can fail to be actualized)
$U(x)$	x is modally invariant across worlds
$N(x)$	x is necessary (actualized at all worlds)
$D(x, y)$	x depends on y (asymmetric modal entailment)
$\Box$	necessarily
$\Diamond$	possibly

Dependence is defined by:

$$D(x, y) \equiv \Box(E(x) \to E(y)) \wedge \neg\Box(E(y) \to E(x)).$$

Formal Conventions

We work in standard constant-domain *S5* with rigid designators. Following Williamson (2013), we distinguish between the logical domain (entities with being) and the existence predicate E (entities that are actualized).

- $\mathcal{D}$ is a fixed domain of all possible individuals (constant across worlds).
- Quantifiers $\forall x, \exists x$ range over $\mathcal{D}$, not over what is actualized at a given world.
- $E(x)$ is a world-relative predicate: $w \models E(x)$ iff x is actualized at w.
- $w \models \varphi$ means that formula φ is true at world w.
- Modal operators are interpreted over a universal accessibility relation on a constant domain.
- All names are rigid: if a term denotes $d \in \mathcal{D}$ at one world, it denotes d at all worlds.

- $D(x, y)$ abbreviates asymmetric modal entailment:

$$D(x, y) \;\equiv\; \Box(E(x) \to E(y)) \;\wedge\; \neg\Box(E(y) \to E(x)).$$

These conventions remain fixed throughout the paper.

3 Axioms and Rationale

Krämer (2024) axiomatizes grounding and modality; MDC shows that, under A0–A8, the uniqueness of an invariant terminus can be derived within constant-domain *S5*.

The axioms governing D (A1–A4) impose structural constraints on dependence, ensuring that it forms an asymmetric, transitive, and well-founded explanatory ordering.

(A4) *Well-foundedness* forbids infinite descending D-chains, ensuring that every non-empty set ordered by D has a D-minimal element (consistent relative to ZFC^{-}). A4 is not introduced to force termination, but to exclude explanatory structures that fail to yield a minimal explanatory endpoint. Since well-foundedness is not first-order expressible, A4 functions as a model-theoretic constraint on the class of admissible MDC models—restricting to structures in which D is well-founded—rather than as an axiom in the proof-theoretic sense.

(A6) *Dependence* requires that every contingent entity directly depends on something invariant:

$$C(x) \to \exists y\,(D(x, y) \wedge U(y)).$$

A6 is best understood not as a bare assumption but as what remains once the alternatives are excluded. A dependence structure must terminate (A4). If the terminus were contingent, its existence would remain unexplained: either the regress restarts, or one accepts a brute stopping point that abandons the intelligibility motivating the framework. A6 formalizes the only remaining option—that the terminus must be something whose existence is not itself in question, namely something invariant. Given transitivity (A3), this ensures that every contingent entity stands in a dependence chain that reaches an invariant element, though the invariant need not be the terminal element of that chain.

A4 and A6 play complementary roles. A4 excludes infinite descending dependence chains, while A6 ensures that contingent dependence is directed toward invariance. Neither suffices alone: A4 without A6 permits termination at contingent endpoints, while A6 without A4 permits infinite chains directed toward invariance. The existence and uniqueness of the Core arise only from their conjunction.

A6 is the signature axiom of MDC: it forces dependence to be invariance-directed, orienting every contingent entity toward what does not vary across worlds. Without it, the Core does not follow, making A6 the central structural commitment of the framework.

(A8) *Terminal uniqueness* is best understood as a framework-relative criterion of individuation internal to MDC rather than as a general metaphysical principle. Suppose two entities are both (i) necessary (N), (ii) invariant (U), and (iii) terminal (admit no further dependence). Such entities share the same modal and structural profile across all worlds: they are actualized at every world, invariant under accessibility, and minimal with respect to the dependence relation.

Within the constant-domain *S5* setting with rigid designation adopted here, the calculus represents entities only through modal and D-relational structure and does not encode haecceitistic or primitive thisness-properties. The calculus deliberately excludes haecceitistic identity conditions, following a structural rather than primitive approach to individuation (cf. Adams 1979). Accordingly, if no modal or D-relational property expressible in the language distinguishes two such entities, MDC has no structural resources by which to individuate them as distinct. In this sense, positing two distinct terminal invariant necessary entities adds no additional structural content.

This does not claim that expressive limits of the language determine identity in reality as such. Rather, it specifies the criterion of individuation adopted within the framework. Uniqueness is therefore not an independent metaphysical claim but a consequence of the adopted criterion of structural individuation. MDC adopts structural individuation as a methodological constraint; alternative identity conditions fall outside its expressive scope.

4 Main Results

Let D^* be the transitive closure of D: $D^*(x, y)$ holds iff $x = y$ or there exists a finite sequence $z_1, \ldots, z_n$ such that

$$D(x, z_1) \wedge D(z_1, z_2) \wedge \cdots \wedge D(z_n, y).$$

This is the reflexive–transitive closure of D.

Theorem 1 (Existence and uniqueness)**.** *There exists a unique* Core $\mathbf{c}$ *such that*

$$N(\mathbf{c}) \wedge U(\mathbf{c}) \wedge \neg\exists z\, D(\mathbf{c}, z).$$

Existence follows from A0, A6, A7, and A4: whenever a contingent entity is actualized, A6 guarantees dependence on something invariant, and A4 ensures that the resulting dependence structure terminates in a minimal element. Uniqueness then follows from A8: any two terminal invariant necessary entities satisfy the same structural profile and therefore coincide. We designate this unique terminus the *Core*, $\mathbf{c}$.

Theorem 2 (Termination)**.** *Every contingent x satisfies*

$$\exists b\, (N(b) \wedge U(b) \wedge D^*(x, b)).$$

Theorem 2 shows that the Core is not merely a logical possibility but the terminus of every contingent dependence chain. No contingent entity is structurally isolated from the invariant endpoint of the calculus.

Theorem 3 (Modal conservation)**.** *If $C(x)$, $D(x, y)$, and $N(y)$, then*

$$(E(x) \to \Box E(y)) \wedge \Diamond \neg E(x).$$

Theorem 3 blocks modal collapse. Although a contingent entity may depend on something necessary, that necessity does not transfer downward. The actualization of x requires y, but the necessity of y does not force x to be actualized in every world.

5 Models

Consistency and interpretability are witnessed by simple constant-domain *S5* models. We work in constant-domain *S5* models $\mathcal{M} = \langle W, R, \mathcal{D}, E \rangle$ with R universal and $\mathcal{D}$ fixed across worlds. Assign one necessary, invariant entity c (*the Core*) and at least one contingent o such that $D(o, c)$ holds at all worlds and $E(o)$ fails at some. All axioms A0–A8 are satisfied: A4 by finitude, A6 by $D(o, c)$, A7 by rigidity of U, and A8 since only c is terminal. Adding further contingent chains $D(p, o)$, $D(q, p)$, etc. preserves all conditions. As a result, MDC admits non-trivial finite models verifying existence, termination, and uniqueness of the Core without modal collapse.

5.1 A Simple Three-World Model

Consider the model with domain $\{c, o, p\}$ and three worlds w_1, w_2, w_3, all mutually accessible. Let c be necessary and invariant, o be contingent and always depend on c, and p be contingent and depend on o.

World	$E(c)$	$E(o)$	$E(p)$
w_1	1	1	1
w_2	1	1	0
w_3	1	0	0

Dependence relations:

$$D(o, c), \qquad D(p, o).$$

This model satisfies A0–A8: c is the only terminal invariant necessary entity, o and p terminate in c, and no modal collapse occurs because $E(o)$ and $E(p)$ fail at some worlds.

6 Objections and Replies

6.1 On Well-Foundedness (A4).

It may be argued that ruling out infinite descending D-chains (A4) merely stipulates a terminus. A4 is not derived but adopted as a minimal constraint on intelligible explanation. MDC treats it as a constraint on explanatory adequacy rather than an arbitrary metaphysical assumption. Competing options carry identifiable costs: (i) *Infinitism* leaves no minimal element and thus no terminating explanation; (ii) *Cycles* violate the asymmetry encoded in A2 and collapse explanatory direction; (iii) *Brute stopping*—a chain terminating at a contingent entity with no further dependence—abandons the intelligibility that motivates the framework. MDC does not refute these options; it shows what follows if explanatory adequacy excludes them.

The well-foundedness condition is standard across grounding-theoretic frameworks and mathematically benign: it is consistent relative to ZFC^{-}. The results of MDC are therefore conditional on A4; any framework that rejects well-foundedness must supply an alternative account of explanatory termination.

6.2 On Invariance and Necessity (A7).

A critic might question whether invariance (U) should entail necessity (N), suggesting that an entity could be invariant across worlds in which it exists while still being contingent.

Reply. Within constant-domain *S5*, the accessibility relation is universal: every world accesses every other. If $U(x)$ holds—meaning that the existence of x does not vary across worlds—then $E(x)$ must hold at all worlds, which is precisely $N(x)$. A7 therefore reflects a consequence of the underlying modal semantics rather than an independent metaphysical commitment.

6.3 On Modal Collapse.

If all contingent entities depend on a necessary Core, a concern arises regarding modal collapse. MDC avoids modal collapse on two independent grounds.

First, the asymmetry of D ensures that dependence is strictly one-directional: $D(x, y)$ does not entail $D(y, x)$, and thus necessity does not transfer from y to x.

Second, A5 independently guarantees that every contingent entity satisfies $\Diamond\neg E(x)$, regardless of what it depends on. Thus, even if a contingent entity depends on something necessary, its own contingency is preserved.

Necessity is inherited by what is depended upon, never by what depends.

6.4 On the Choice of Modal Framework.

One might object that the use of constant-domain *S5* with rigid designation renders the results framework-dependent, and that alternative semantics could undermine the uniqueness of the Core.

Reply. Constant-domain *S5* is the natural setting for MDC precisely because invariance requires existence to be stable across worlds. Varying-domain or non-rigid semantics, by allowing entities to exist in some worlds but not others without constraint, weaken or eliminate the notion of invariance formalized by $U(x)$.

The results of MDC are therefore conditional on the adopted semantics, but they are non-trivial consequences of the axioms within that framework, not artifacts of the semantic choice. MDC does not claim invariance must be modeled in S5; it shows what follows when invariance is represented within this standard modal framework. More broadly, MDC is conditional not only on axioms A0–A8 but on the adequacy of the underlying Kripkean modal framework as a representation of modal structure; traditions that reject such a framework will thereby reject the applicability of the results rather than their internal validity.

6.5 On Uniqueness and Expressive Limits (A8).

A critic may view A8 as an arbitrary enforcement of monism. The charge deserves a careful answer, which proceeds in two steps: first by structural exhaustion, then by grounding the individuation criterion in MDC's explanatory aims.

Structural Exhaustion. The uniqueness condition follows from an exhaustion of cases that requires no appeal to the individuation criterion at all. Suppose b_1 and b_2 are both terminal, invariant, and necessary. There are exactly two cases.

Case 1: $D(b_1, b_2)$ or $D(b_2, b_1)$ holds. Then one entity depends on the other and is therefore not terminal—contradicting the assumption.

Case 2: Neither $D(b_1, b_2)$ nor $D(b_2, b_1)$ holds. Then b_1 and b_2 agree on every modal property (N, U, E at every world) and stand in no D-relational difference expressible in the language of MDC. Within this framework, they are structurally indiscernible.

These two cases are jointly exhaustive and mutually exclusive. Any purported pair of terminal invariant necessary entities either violates terminality or collapses into indiscernibility. There is no third case.

Why Structural Individuation Is Not Arbitrary. A critic may grant the exhaustion argument while objecting that structural indiscernibility does not entail identity—that MDC simply *refuses* the resources needed to distinguish such entities and thereby rigs the result. This objection deserves a direct reply.

The adoption of structural individuation is not an arbitrary restriction but reflects the explanatory aim of the framework. MDC is concerned with dependence relations as they are expressed through modal and D-relational structure. Introducing primitive identity conditions that generate no difference in any modal or dependence relation expressible within MDC would add distinctions that are, within the framework's expressive scope, entirely idle—they play no role in any dependence, modal, or explanatory relation the calculus can represent. Far from supplementing MDC, such additions would reintroduce precisely the kind of ungrounded multiplicity the framework is designed to diagnose.

Within constant-domain *S5* with rigid designation, MDC individuates entities solely through modal and D-relational structure (cf. Adams 1979). This is not a claim that expressive limits determine identity in reality as such; it is the constitutive criterion of individuation adopted by the framework. Alternative identity conditions fall outside its expressive scope, and any framework that requires them must explain what explanatory work they do that MDC's structural resources cannot.

6.6 On Hyperintensionality.

A critic may object that grounding is hyperintensional, whereas D is defined modally, and therefore MDC fails to capture grounding properly.

Reply. MDC is a modal surrogate for grounding, not an equivalent. $D(x, y)$ captures grounding-like structure along three dimensions: *asymmetry* (non-reciprocity of D, encoded in A2), *directionality* (priority of y over x in the dependence ordering), and *necessitation* ($\Box(E(x) \to E(y))$ ensures that actualization of x requires actualization of y). What D does not capture is hyperintensional discrimination between necessarily equivalent propositions. This is a deliberate limitation, not an oversight.

MDC shows how far these structural features of grounding can be recovered within standard modal resources. It does not claim that modal entailment exhausts the content of metaphysical grounding. Richer hyperintensional accounts address different explanatory goals; MDC operates at the level of modal structure and derives what follows there.

Accordingly, objections to MDC target either the adopted constraints or the adequacy of modal resources, not the internal validity of the derivations.

7 Consequences and Extensions

Positive Yield of the Framework. Before developing specific consequences, it is worth stating explicitly what MDC delivers as a positive result.

First, MDC supplies a principled explanatory terminus for contingent dependence without invoking theology, primitive causation, or hyperintensional grounding operators. The Core is derived from structural constraints alone.

Second, MDC functions as a constraint on any adequate theory of dependence. Any causal, grounding-theoretic, or scientific account that satisfies A0–A8 inherits the Core result. MDC does not compete with such theories; it delimits the structural space within which they operate.

Third, structural monism is not assumed but derived. Classical monist frameworks—from Aristotle to Spinoza—posited a unique necessary ground as a starting commitment. MDC shows that this commitment follows from minimal structural conditions on dependence, invariance, and individuation. Monism is the structurally minimal outcome given the adopted constraints.

MDC derives uniqueness of an invariant terminus within plain quantified *S5* without invoking a primitive grounding operator. This constitutes an economy result *relative to that specific target*: existing frameworks such as Fine (2012), Litland (2017), and Krämer (2024) deploy hyperintensional machinery in pursuit of a richer explanandum that MDC does not attempt to reproduce. The comparison is not one of overall theoretical power but of resources deployed relative to the structural result achieved.

The contribution is not merely internal to the system: any account of ontological dependence that satisfies A0–A8 inherits the uniqueness result, making MDC a structural constraint on competing theories insofar as they satisfy A0–A8, rather than merely one proposal among others.

The calculus yields three formal consequences that extend grounding theory into the modal domain.

Interpretive Consequence (Essence and Grounding). If $D^*(x, \mathbf{c})$ and $N(\mathbf{c}) \wedge U(\mathbf{c})$, then it is natural to interpret x as essentially grounded in $\mathbf{c}$ in a Finean sense, since no world contains x without the Core. This connection is interpretive rather than derived within MDC itself.

Corollary 1 (Essential Ground Uniqueness)**.** *If x is essentially grounded in b and b satisfies $N(b) \wedge U(b)$, then $b = \mathbf{c}$.*

Under this interpretation, essence admits a single invariant terminus: all essential grounds coincide in the Core.

Theorem 4 (Uniqueness of Origin within Models)**.** *In any constant-domain S5 model satisfying A0–A8, there is exactly one object b such that $N(b)$ and $\forall x\,(C(x) \to D^*(x, b))$.*

Sketch. It follows directly from Theorems 1–2 and A8: every contingent chain terminates in a single necessary, invariant b that exists at all worlds. □

The uniqueness established here is strictly intra-model: within any constant-domain *S5* model satisfying A0–A8, there is exactly one such object b. MDC does not assert numerical identity of the Core across distinct models; establishing cross-model identity would require additional identity conditions beyond the structural resources of the calculus.

Theorem 5 (Essence from Invariance)**.** $N(x) \wedge U(x) \to \Box\forall y\,(E(y) \to E(x))$.

In particular, for $x = \mathbf{c}$, the Core is actualized in every world where anything exists, and its necessity is established independently by Theorem 1.

Corollary 2 (Structural Monism)**.**

$$\Box\exists x\, E(x) \;\rightarrow\; \big(E(\mathbf{c}) \wedge \forall y\, ((N(y) \wedge U(y) \wedge \forall z\, \neg D(y,z)) \rightarrow y = \mathbf{c})\big).$$

Proof. Assume $\Box\exists x\, E(x)$. By Theorem 1, there exists a unique object $\mathbf{c}$ such that

$$N(\mathbf{c}) \wedge U(\mathbf{c}) \wedge \forall z\, \neg D(\mathbf{c}, z).$$

Hence $N(\mathbf{c})$, and therefore $E(\mathbf{c})$.

Now let y be arbitrary and suppose

$$N(y) \wedge U(y) \wedge \forall z\, \neg D(y, z).$$

Then y satisfies the same terminal invariant necessary profile as $\mathbf{c}$. By A8, it follows that $y = \mathbf{c}$.

Therefore

$$E(\mathbf{c}) \wedge \forall y\, ((N(y) \wedge U(y) \wedge \forall z\, \neg D(y,z)) \rightarrow y = \mathbf{c}),$$

as required. □

If any world contains being, then the Core is actualized, and any necessary invariant terminal entity coincides with it. In this sense, a form of structural monism emerges as a theorem of modal dependence rather than as an independent assumption.

The structure established by Corollary 2 bears comparison to classical monist frameworks, including Spinoza's substance monism, in which all contingent modes depend on a single necessary substance. MDC arrives at an analogous result through independent modal constraints rather than metaphysical stipulation. The structural role of the Core bears comparison not only to Western monist traditions but also to non-dual frameworks such as Advaita Vedānta, though MDC itself remains formally non-committal on metaphysical interpretation.

8 Conclusion

The Modal–Dependence Calculus shows that a minimal set of structural constraints on asymmetric modal entailment suffices to recover a structural metaphysical picture traditionally associated with grounding theory and classical monism.

By combining axioms governing contingency, necessity, invariance, and well-founded dependence, MDC establishes three results: that contingent entities exist, that all dependence chains terminate, and that the terminal point of such chains is unique.

These results are derived within a constant-domain *S5* system using a single primitive dependence relation. The calculus does not presuppose a substantive metaphysical account of grounding or causation, but specifies the structural conditions under which such accounts must operate if they are to be coherent.

In this respect, MDC is not a theory of how dependence operates, but a framework for determining what must be true of any such operation. It identifies the minimal constraints required for a dependence structure to be intelligible: asymmetry, access to invariance, and termination. The resulting picture is one in which variation is conditioned by invariance and the structure of dependence is oriented toward a unique terminus.

MDC thus provides a model of structural individuation within a modal framework. It does not establish that such individuation exhausts the metaphysics of identity, but shows how dependence and termination behave under that criterion.

The structural criterion of individuation adopted here reflects a broader commitment to the view that identity is exhausted by modal and relational structure. The results of MDC do not presuppose the defense of that commitment; the defense is developed independently in subsequent work.

On Disagreement and Constraint. Disagreement about the interpretation of dependence, the nature of invariance, or the status of the terminal point remains possible. MDC does not resolve such disagreements by stipulation, but constrains them: any account that violates these structural conditions must explain how coherent dependence remains possible without them. The calculus therefore does not settle metaphysical questions about the nature of reality, but delineates the space within which such questions can be meaningfully posed.

The results are conditional on A0–A8 and the constant-domain *S5* framework. MDC inherits the undecidability of quantified modal logic; decidability results for fragments, completeness theorems, and proof-theoretic development remain future work. Extensions to weaker modal systems and investigation of the categorical structure of MDC models are further open questions.

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The following appendices collect the full axiom list, formal derivations of the principal theorems, further model constructions, an intuitive overview, worked examples, and a dialectical map of structural alternatives.

A Full Axiom List

A0	$\exists x\, C(x)$ (Contingency)
A1	$\neg D(x,x)$ (Irreflexivity)
A2	$D(x,y) \to \neg D(y,x)$ (Asymmetry)
A3	$(D(x,y) \wedge D(y,z)) \to D(x,z)$ (Transitivity)
A4	No infinite descending D-chain (Well-foundedness)
A5	$C(x) \to \Diamond \neg E(x)$ (Contingency preserves $\Diamond\neg E$)
A6	$C(x) \to \exists y\, (D(x,y) \wedge U(y))$ (Direct dependence on invariant)
A7	$U(x) \to N(x)$ (Invariance suffices for necessity)
A8	$\big([\forall z\, \neg D(x,z) \wedge U(x) \wedge N(x)] \wedge [\forall z\, \neg D(y,z) \wedge U(y) \wedge N(y)]\big) \to x = y$ (Modal–relational indiscernibility entails identity)

Note on axiom roles. A4 and A6 are jointly sufficient for termination: A4 forbids infinite descent while A6 ensures chains are directed toward invariance; neither suffices alone. A5 is independent of both and functions solely as the collapse-blocker. A8 is independent of A4 and A6: it governs uniqueness of the terminus but plays no role in establishing that a terminus exists. A7 connects the predicate families U and N and is required for the invariance-to-necessity inference; removing it would decouple these two properties. A1–A3 are standard partial-order constraints on D and are mutually independent.

B Proofs of the Main Theorems

Preliminaries. We work throughout in constant-domain *S5* with rigid designators. Modal reasoning uses standard rules: Necessitation, Duality, K, and S5's axioms (T, 4, 5). D^* denotes the reflexive–transitive closure of D.

The proofs that follow are model-theoretic existence arguments conducted in set-theoretic metatheory, not derivations within the object-level system MDC. The theorems establish that every MDC-satisfying model contains a Core; they do not assert derivability from A0–A8 by proof rules alone.

Theorem 1 (Existence and Uniqueness of the Core).

Proof. By A0, $\exists x\, C(x)$. Fix a such that $C(a)$.

By A6, there exists b_0 such that $D(a,b_0) \wedge U(b_0)$. Hence $D^*(a,b_0)$, and by A7, $N(b_0)$.

Define $S = \{\, y \mid D^*(a,y) \wedge U(y) \,\}$. Then S is nonempty since $b_0 \in S$.

By A4 (well-foundedness of D), every nonempty subset of $\mathcal{D}$ ordered by the strict partial order D has a D-minimal element; note that S is here ordered by D rather than D^*, so A4 applies directly. Hence there exists $m \in S$ such that $\forall z\, \neg D(m, z)$. Since $m \in S$, we have $U(m)$ and $D^*(a, m)$; by A7, $N(m)$.

Thus $\exists b\, (N(b) \wedge U(b) \wedge \forall z\, \neg D(b, z))$.

For uniqueness, suppose b_1 and b_2 both satisfy $N(b_i) \wedge U(b_i) \wedge \forall z\, \neg D(b_i, z)$ for $i = 1, 2$. The antecedent of A8 holds with $x := b_1$ and $y := b_2$. By A8, $b_1 = b_2$.

Therefore $\exists!\, b\, (N(b) \wedge U(b) \wedge \forall z\, \neg D(b, z))$. □

Theorem 2 (Termination).

Proof. Let x be arbitrary with $C(x)$. By A6, choose b_0 with $D(x, b_0)$ and $U(b_0)$; by A7, $N(b_0)$.

Define $S = \{\, y \mid D^*(x, y) \wedge U(y) \,\}$, nonempty since $b_0 \in S$. By A4 (well-foundedness of D), S ordered by D has a D-minimal element m with $\forall z\, \neg D(m, z)$. Since $m \in S$: $D^*(x, m)$, $U(m)$, and by A7, $N(m)$.

Thus $\exists b\, (N(b) \wedge U(b) \wedge D^*(x, b))$. □

Theorem 3 (Modal Conservation).

Proof. Assume $C(x)$, $D(x, y)$, and $N(y)$. From $D(x, y)$: $\Box(E(x) \to E(y)) \wedge \neg\Box(E(y) \to E(x))$. From $N(y)$: $\Box E(y)$, so $E(x) \to \Box E(y)$. From A5 and $C(x)$: $\Diamond\neg E(x)$. Therefore $(E(x) \to \Box E(y)) \wedge \Diamond\neg E(x)$. □

C Additional Models

C.1 Three-World Canonical Model

Consider a constant-domain *S5* model with domain $\{c, o, p\}$ and three worlds w_1, w_2, w_3, all mutually accessible. The table records the extension of E, U, and N at each world; invariance and necessity assignments are constant across worlds, reflecting the rigidity of U and N in constant-domain *S5*.

World	$E(c)$	$E(o)$	$E(p)$	$U(c)$	$U(o)$	$U(p)$	$N(c)$	$N(o)$	$N(p)$
w_1	1	1	1	1	0	0	1	0	0
w_2	1	1	0	1	0	0	1	0	0
w_3	1	0	0	1	0	0	1	0	0

Dependence relations (global): $D(o, c)$, $D(p, o)$.

In this model, c is invariant and necessary while o and p are contingent and fail to be actualized at some worlds. All axioms A0–A8 are satisfied: dependence is well-founded, all chains terminate in c, and no modal collapse occurs since $E(o)$ and $E(p)$ fail at w_3 and w_2–w_3 respectively.

C.2 Extended Chains

MDC admits arbitrarily long but finite dependence chains. For any sequence

$$x_n \to x_{n-1} \to \cdots \to x_1 \to c,$$

where each x_i is contingent and c satisfies $U(c) \wedge N(c)$, all axioms A0–A8 remain satisfied. Each element in the chain depends directly or transitively on c, and by A4 no chain can be infinitely descending. Dependence may thus be extended without bound in length, provided each individual chain is finite.

C.3 Branching Models

MDC also permits branching dependence structures. Let

$$p \to o_1 \to c \qquad \text{and} \qquad q \to o_2 \to c,$$

where o_1 and o_2 are contingent and c is the unique terminal invariant necessary entity. Multiple independent chains may converge on the same terminus without violating any axiom. By A8, no branching structure can yield more than one terminal invariant necessary entity: any two such candidates would be structurally indiscernible and therefore identical.

C.4 Infinite Trees with Finite Depth

MDC admits models with infinitely many contingent entities arranged in a branching dependence structure, provided that all dependence chains are finite. Specifically, there may be countably many distinct branches, each of finite length terminating at c, without violating A4.

A4 excludes infinite *descending* chains of dependence (infinite depth) but places no restriction on the *number* of such chains (breadth). Accordingly, MDC permits infinite but well-founded universes in which every contingent entity stands in a finite dependence relation ultimately terminating in c.

MDC therefore allows ontological abundance without explanatory regress.

D Intuitive Overview of MDC

D.1 Why We Need a Terminus

Contingent things can fail to be actualized and cannot be their own explanation. If something is contingent, there must be something it depends on. The question is how such dependence can be

structured.

D.2 What Dependence Really Means

$D(x, y) \equiv \Box(E(x) \rightarrow E(y)) \wedge \neg\Box(E(y) \rightarrow E(x))$: if x is actualized, y must be; but y may be actualized without x.

D.3 Why Chains Must End

There are only three possibilities for a dependence chain:

- It continues infinitely
- It loops back on itself
- It terminates

Infinite chains never reach an explanation, and cycles violate asymmetry. So dependence must terminate.

D.4 Why the Terminus Is Invariant

MDC requires that contingent entities ultimately depend on something that does not vary across possible worlds. This invariant element provides a stable endpoint for dependence.

D.5 Why the Terminus Is Unique

Two invariant, necessary, terminal entities would share the same modal and dependence structure. Under structural individuation, there is no basis for distinguishing them. So they coincide.

D.6 The Core and Modal Stability

The Core is actualized in every world in which anything is actualized.

D.7 No Modal Collapse

$\Diamond\neg E(x)$ holds for every contingent x. The Core does not determine which contingent entities are actualized.

D.8 Final Summary

- Dependence = asymmetric modal entailment.
- Chains of dependence cannot be infinite or cyclical.
- They must terminate in an invariant, necessary endpoint.
- That endpoint is unique.

- Structural monism emerges as a theorem, not an assumption.

E Worked Examples

E.1 A Simple Two-Object Model

Domain $\{c, o\}$, three worlds. $D(o, c)$. c invariant, necessary, terminal. o contingent, fails at w_2, w_3. Core $= c$.

E.2 A Three-Level Dependence Chain

$D(o_2, o_1)$, $D(o_1, c)$. Both o_2 and o_1 terminate in the single Core c via A6 and A4.

E.3 Modal Verification of D

$\Box(E(o_1) \to E(c))$ holds at every world; $\neg\Box(E(c) \to E(o_1))$ holds since $E(c)$ holds at w_3 while $E(o_1)$ fails. So $D(o_1, c)$ is verified.

E.4 Branching Without Multiplying Terminators

$p \to o_1 \to c$ and $q \to o_2 \to c$. Both branches terminate in c. A8 prevents two distinct terminals.

E.5 Attempted Countermodel Fails

Two terminals c_1, c_2 both satisfying N, U, terminal. A8 forces $c_1 = c_2$.

E.6 A Model Avoiding Modal Collapse

$D(o, c)$, $N(c)$, $C(o)$. At w_1: $E(o) = 1$. At w_2: $E(o) = 0$. So $\Diamond\neg E(o)$ and $\Box E(c)$. No collapse.

E.7 Infinite Universe with Finite Depth

Infinitely many contingents $p_1, p_2, \ldots$, each branch $p_n \to \cdots \to p_1 \to c$ finite. A4 forbids infinite depth. All chains terminate at c.

F Structural Displacement of Plurality: A Dialectical Map

This appendix maps the structural costs of maintaining plurality within MDC. It is not a refutation of pluralism as a metaphysical position, but a precise account of which axioms must be rejected or supplemented for plurality to be sustained.

The core result. MDC does not eliminate pluralism; it shows that maintaining plurality requires abandoning one or more of the structural conditions that generate explanatory coherence within the framework.

Cost map.

Move	Structural Cost
Reject A8	Must accept haecceitism: primitive thisness as an identity condition not capturable by modal or D-relational structure (cf. Adams 1979).
Accept A8 but differentiate terminals	Must introduce properties that generate no modal or D-relational differences. Such properties introduce distinctions not captured by the structural resources of the framework.
Reject A6	The framework no longer guarantees that contingent dependence is directed toward invariance. An alternative account of how termination occurs without invariant endpoints is then required.
Reject A4	Must accept infinitism about explanation: that contingent dependence chains need not terminate. This is a coherent but substantive commitment.

What this shows. Each move is a legitimate philosophical option. MDC does not claim otherwise. What it shows is that plurality is not a default position that MDC fails to rule out—it is a position that requires positive theoretical additions or deletions. The framework makes those costs explicit and locates them precisely.

The asymmetry between monism and plurality. Monism within MDC requires no commitments beyond A0–A8. Plurality requires either additional primitive identity conditions, additional non-structural properties, or the rejection of well-foundedness or invariant-directed dependence. In this sense monism is the structurally minimal outcome: it is what the framework derives when no supplementary commitments are made.

One-line summary. MDC does not resolve the monism–pluralism dispute by decree. It clarifies what must be assumed, rejected, or added for that dispute to be sustained on the pluralist side.

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